

DATA 260 — HOMEWORK 3**Authenticated Rental-Housing Portal and Retrieval Comparison**

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1. Configuration and Reproducibility

This submission extends the shared DATA-260 repository and keeps the rental-housing application code at the repository root while placing HW3 evidence under reports/hw03/.

Github Link - <https://github.com/pragyachourasia-05/data260-6758/tree/hw3>

Configuration item	Value
SID4	6758
PORT_BASE	8458
PREFIX	s6758
SEED	6758
VERIFY_SEED	266758
DOMAIN_ID / domain	6 / Rental housing listings
Embedding model	sentence-transformers/all-MiniLM-L6-v2
Local application port	http://localhost:8458
Final tagged commit	4201a060c11c1ddf2e4c358acf875cc53f074dc6
Hardware	Windows laptop; 13th Gen Intel Core i5-1335U; 10 cores; 12 logical processors; 15.69 GB RAM; CPU execution
Local model	llama3.2:3b
GitHub repository	https://github.com/pragyachourasia-05/data260-6758/tree/hw3

I worked in my virtual environment, installed the project dependencies, and started the application with main.py. I opened the portal at <http://localhost:8458> and checked the home page, login page, dashboard, invalid-login message, logout behavior, and session-cookie details. I then ran my retrieval comparison with $k=3$. The run produced retrieval_results.json and retrieval_summary.json in reports/hw03/raw, which I used for the tables and discussion in this report.

For my corpus, I used the California tenant guide and the HUD fair-housing guide because both are relevant to rental housing. I recorded their filenames and SHA-256 hashes in CORPUS_MANIFEST.json. I also tried to download the San José rent-stabilization page, but the site returned Access Denied. I therefore left that page out instead of treating an incomplete download as evidence.

2. Overview

In this homework, I extended my rental-listing application in two ways. I added session-based authentication with a protected dashboard and logout. I also compared three ways of splitting my rental-housing documents before retrieval. I kept the experiment retrieval-only so that I could evaluate the returned passages directly.

3. Part 1 — Authentication and Session Security

The home page links to login. A successful admin/password submission stores the username in the signed Starlette session and redirects to /dashboard. The dashboard checks the session before rendering. Logout clears the session and returns the browser to the home page. Invalid credentials redirect back to the login page with an error alert.

- Protected route: /dashboard redirects unauthenticated users to /login.
- Idle timeout: the authentication router rejects sessions that exceed the configured idle interval.
- Cookie controls: SessionMiddleware sets Secure, HttpOnly, SameSite=Lax, and a one-hour maximum age.

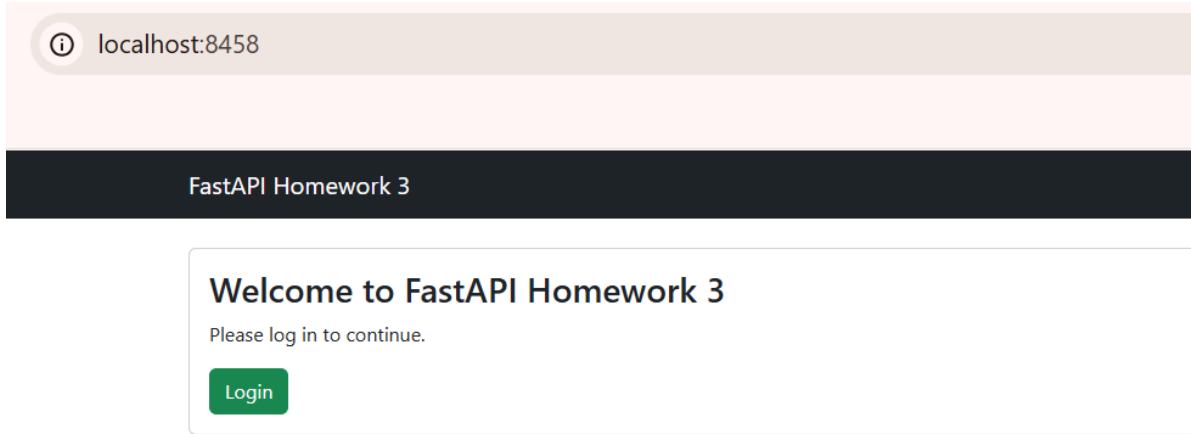


Figure 1. Home page while logged out.

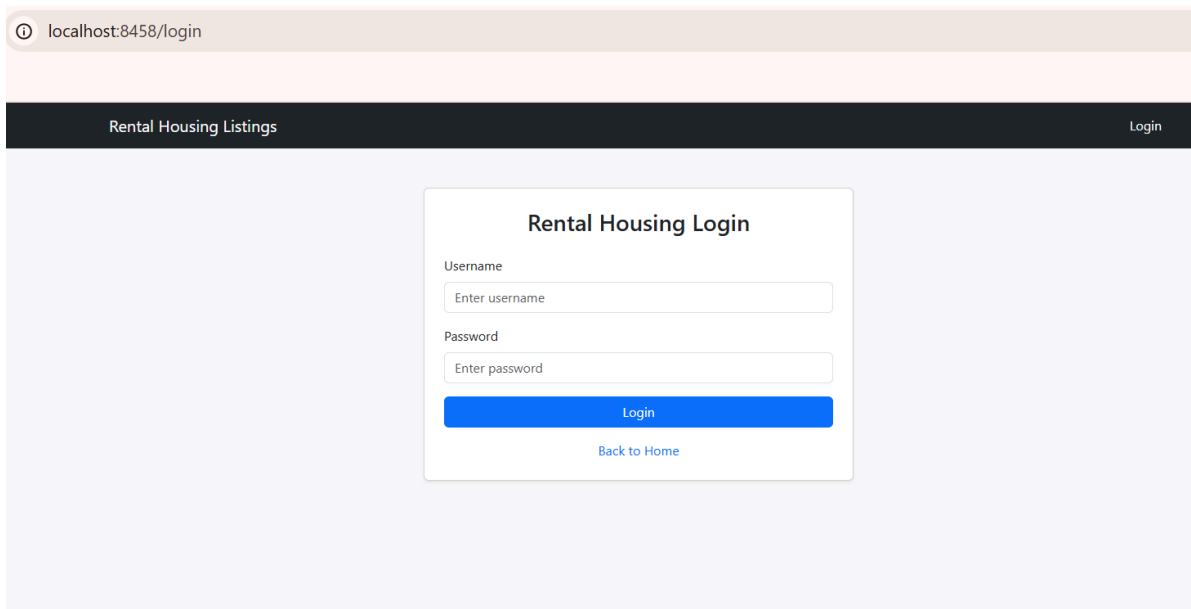


Figure 2. Login form.

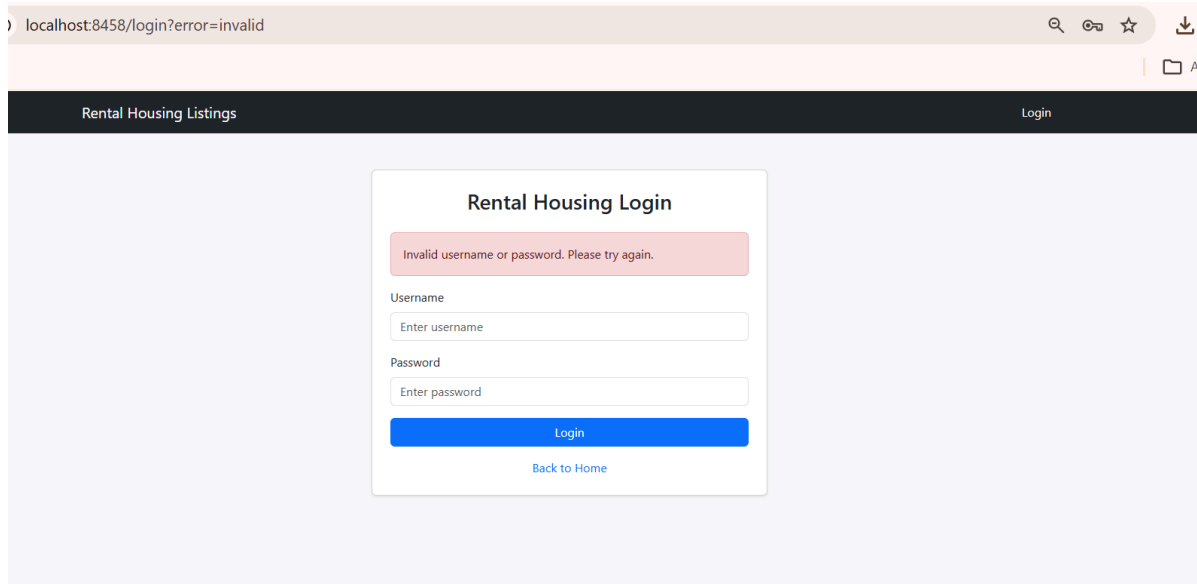


Figure 3. Invalid-credential alert.

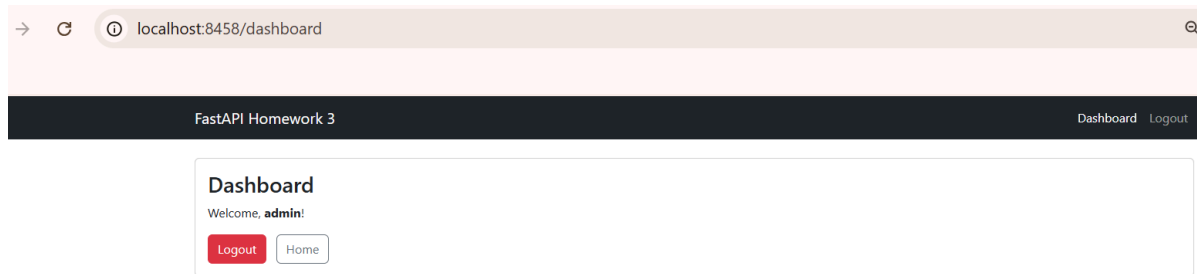


Figure 4. Authenticated dashboard.

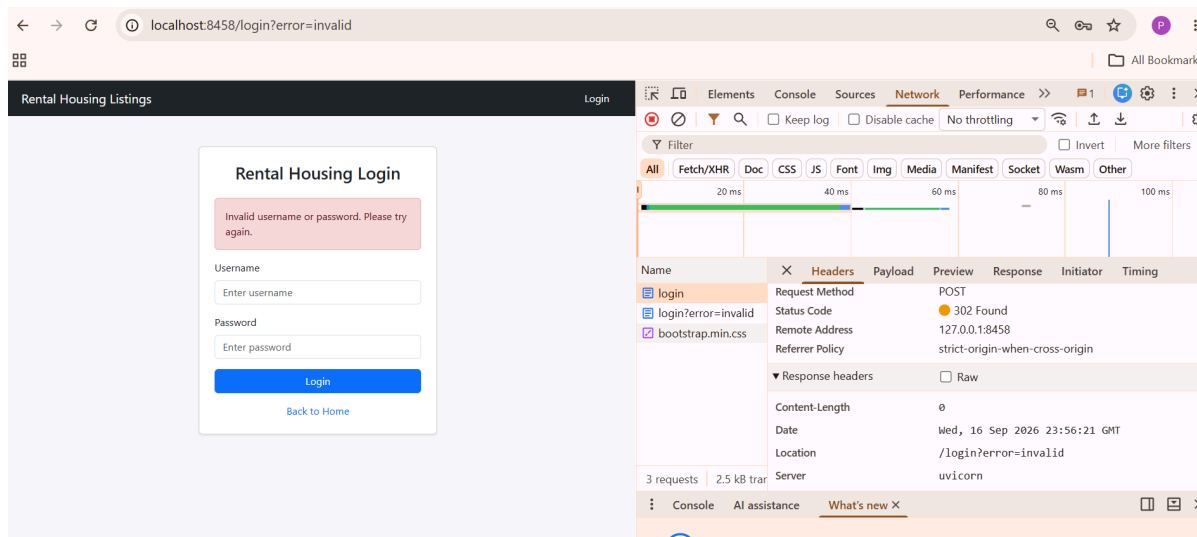


Figure 5. Browser network evidence for the authentication request and session cookie.

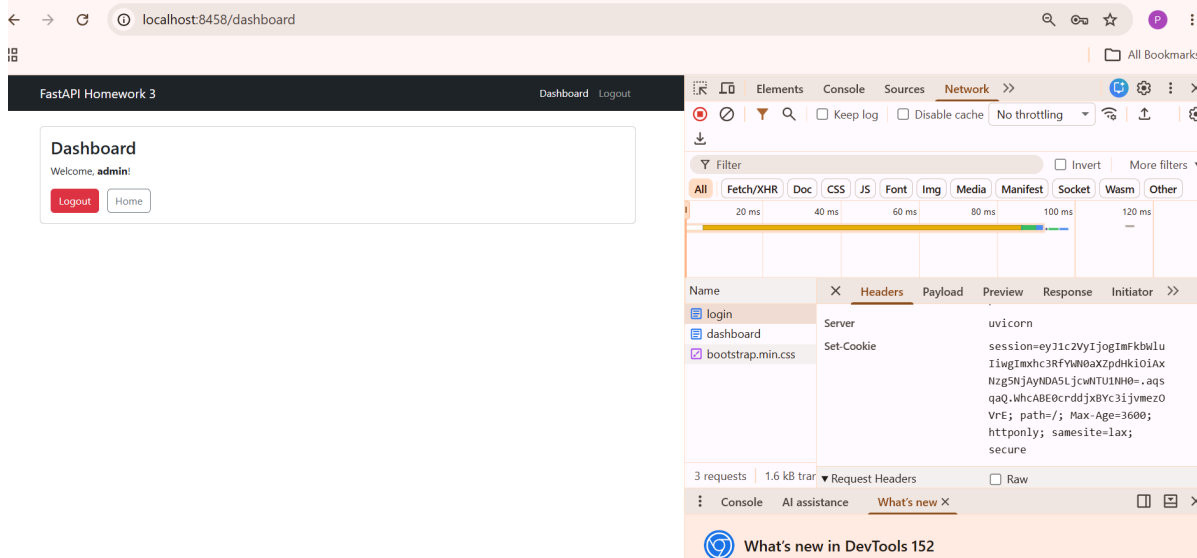


Figure 6. Dashboard/network evidence after authentication.

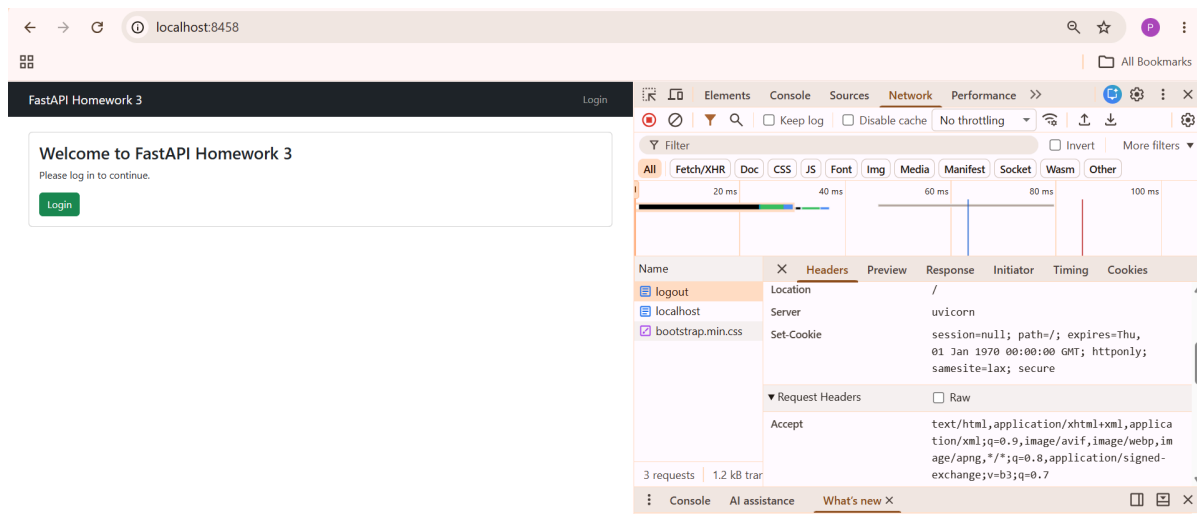


Figure 7. Logged-out home/network state.

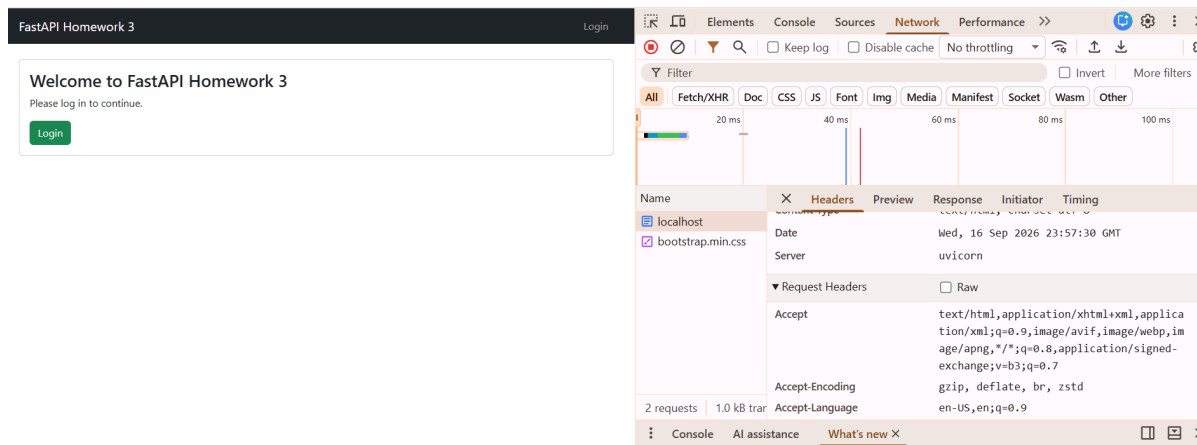


Figure 8. Home page with browser network panel.

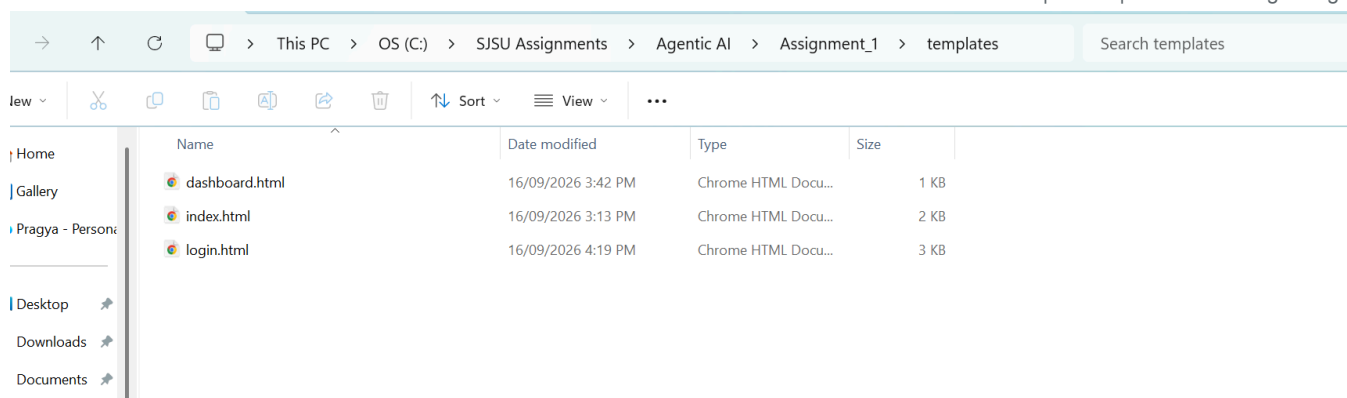


Figure 9. Template files used for the home, login, and dashboard pages.

4. Part 2 — Retrieval Experiment

Five questions were stored in questions.yaml. They cover Fair Housing Act protections, prohibited conduct, disability accommodations, California entry notice, and security-deposit timing. Each result records the query vector dimension and first eight values, the retrieved rank, vector-store score, independently computed cosine similarity, chunk length, preview text, source filename, source-level Recall@3, and retrieval latency.

4.1 Chunking methods

- Token splitter: fixed-size chunks with overlap.
- Semantic splitter: boundaries selected using embedding-change percentiles.
- Sentence-window splitter: sentence-centered nodes with nearby context in metadata.

4.2 Token and semantic output

```
===== TECHNIQUE: TOKEN =====
Chunks created: 88
Building vector index...

Running question q1: What does the Fair Housing Act protect renters from?
Query vector dimension: 384
Query vector first 8: [-0.05613279715180397, 0.025396961718797684, 0.035968441516160965, 0.02243846468627453, 0.0347027
0335674286, 0.07276196777820587, 0.014962323941290379, -0.07519298046827316]
Document vector shape: 3 x 384
Retrieval latency: 94.11 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.6497226045321837 | 0.6521 | 2760 | FAIR HOUSING Equal Opportunity for All U.S. Department of Housing and Urban
Development Office of Fair Housing and Equal Opportunity Please visit our website: w
2 | 0.600499971799957 | 0.4461 | 2273 | any statement, in connection with the provision of homeowners insurance cov
erage, that indicates a preference, limitation or discrimination based on race, col
3 | 0.5943235032872336 | 0.6374 | 2205 | Happens After A Complaint Investigation?.....12
In Addition.....12

Running question q2: Which groups are protected by the Fair Housing Act?
Query vector dimension: 384
Query vector first 8: [0.009971993044018745, -0.02650557830929756, 0.032666221261024475, 0.023070378229022026, 0.065730
07255792618, 0.05767792463302612, 0.020607110112905502, -0.09983338415622711]
Document vector shape: 3 x 384
Retrieval latency: 41.60 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.6737072597624503 | 0.6594 | 2760 | FAIR HOUSING Equal Opportunity for All U.S. Department of Housing and Urban
Development Office of Fair Housing and Equal Opportunity Please visit our website: w
2 | 0.6309321243906894 | 0.6691 | 2205 | Happens After A Complaint Investigation?.....12
```

Figure 9. Token-chunk retrieval output, including vector dimension, shape, scores, cosine values, and previews.

```

Running question q3: Can a landlord discriminate against a renter with a disability?
Query vector dimension: 384
Query vector first 8: [-0.018441710621118546, 0.08497778326272964, -0.05406013876199722, 0.03909369185566902, -0.030797
790735960007, 0.0481431670486927, 0.028272682800889015, -0.05153852701187134]
Document vector shape: 3 x 384
Retrieval latency: 160.19 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.5912427312050823 | 0.6178 | 2273 | any statement, in connection with the provision of homeowners insurance co
verage, that indicates a preference, limitation or discrimination based on race, col
2 | 0.5520202943063943 | 0.5964 | 2433 | does not accept you as a tenant, the landlord must return the entire holdin
g deposit. UNLAWFUL DISCRIMINATION What is unlawful discrimination? A landlord ca
3 | 0.520357262265131 | 0.4984 | 2424 | survivor of domestic violence, because this may be an example of sex discri
mination under both California and Federal law. These laws are subject to the limit

Running question q4: How long does a California landlord generally have to return a security deposit?
Query vector dimension: 384
Query vector first 8: [0.08526249974966049, 0.01604536548256874, -0.0038037181366235018, 0.06782228499650955, -0.023465
050384402275, 0.00406623724848032, 0.0025624465197324753, -0.09909654408693314]
Document vector shape: 3 x 384
Retrieval latency: 273.94 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.6762569673747947 | 0.5864 | 2298 | Toolkit-Jan-2016updated.pdf. Normal security deposit law applies in cases
where a survivor has to break their lease early.141 In addition, if the tenant is a
2 | 0.6667704119087228 | 0.5841 | 2349 | purpose, as described on pages 74-76. Almost all landlords charge tenants
a security deposit. The security deposit may be called last month's rent, security d
3 | 0.6620268202620734 | 0.5371 | 2257 | (For example, 30 days advance written notice normally is required in a mo
nth-to-month rental agreement.). In a fixed-term rental agreement, a change in depos

```

Figure 10. Additional token queries and a visible high-scoring but weakly targeted retrieval.

```

===== TECHNIQUE: SEMANTIC =====
Chunks created: 51
Building vector index...

Running question q1: What does the Fair Housing Act protect renters from?
Query vector dimension: 384
Query vector first 8: [-0.05613279715180397, 0.025396961718797684, 0.035968441516160965, 0.02243846468627453, 0.0347027
0335674286, 0.07276196777820587, 0.014962323941290379, -0.07519298046827316]
Document vector shape: 3 x 384
Retrieval latency: 19.63 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.691193267108845 | 0.6466 | 9717 | 20410-2000 i FAIR HOUSING Equal Opportunity for All ii FAIR HOUSING Equal
Opportunity for All The Fair Housing Act The Fair Housing Act prohibits discrim
2 | 0.6497226045321837 | 0.6521 | 2622 | FAIR HOUSING Equal Opportunity for All U.S. Department of Housing and Urban
Development Office of Fair Housing and Equal Opportunity Please visit our website: w
3 | 0.5958197978945887 | 0.3296 | 39 | Department of Justice for enforcement.

Running question q2: Which groups are protected by the Fair Housing Act?
Query vector dimension: 384
Query vector first 8: [0.009971993044018745, -0.02650557830929756, 0.032666221261024475, 0.023070378229022026, 0.065730
07255792618, 0.05767792463302612, 0.020607110112905502, -0.09983338415622711]
Document vector shape: 3 x 384
Retrieval latency: 19.13 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.7269476829219434 | 0.6892 | 9717 | 20410-2000 i FAIR HOUSING Equal Opportunity for All ii FAIR HOUSING Equal
Opportunity for All The Fair Housing Act The Fair Housing Act prohibits discrim
2 | 0.6737072597624503 | 0.6594 | 2622 | FAIR HOUSING Equal Opportunity for All U.S. Department of Housing and Urban
Development Office of Fair Housing and Equal Opportunity Please visit our website: w
3 | 0.58935433561629 | 0.3527 | 39 | Department of Justice for enforcement.

```

Figure 11. Semantic-chunk retrieval output.

4.3 Sentence-window output

```
===== TECHNIQUE: SENTENCE_WINDOW =====
Chunks created: 951
Building vector index...

Running question q1: What does the Fair Housing Act protect renters from?
Query vector dimension: 384
Query vector first 8: [-0.05613279715180397, 0.025396961718797684, 0.035968441516160965, 0.02243846468627453, 0.0347027
0335674286, 0.07276196777820587, 0.014962323941290379, -0.07519298046827316]
Document vector shape: 3 x 384
Retrieval latency: 44.91 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.6864758007081394 | 0.7742 | 42 | The Fair Housing Act covers most housing.
2 | 0.6831743716133692 | 0.6764 | 200 | 3 FAIR HOUSING Equal Opportunity for All However, the Fair Housing Act
does not protect a person who is a direct threat to the health or safety of others
3 | 0.6800432896342086 | 0.6793 | 333 | The Department of Housing and Urban Development (HUD) enforces the Fair Hou
sing Act, which prohibits discrimination and the intimidation of people in their ho

Running question q2: Which groups are protected by the Fair Housing Act?
Query vector dimension: 384
Query vector first 8: [0.009971993044018745, -0.02650557830929756, 0.032666221261024475, 0.023070378229022026, 0.065730
07255792618, 0.05767792463302612, 0.020607110112905502, -0.09983338415622711]
Document vector shape: 3 x 384
Retrieval latency: 41.14 ms
rank | store_score | cosine_sim | chunk_len | preview
1 | 0.7099055382542925 | 0.7229 | 200 | 3 FAIR HOUSING Equal Opportunity for All However, the Fair Housing Act
does not protect a person who is a direct threat to the health or safety of others
2 | 0.6994555863260808 | 0.8027 | 42 | The Fair Housing Act covers most housing.
3 | 0.6848143543859755 | 0.6646 | 453 | 20410-2000 i FAIR HOUSING Equal Opportunity for All ii FAIR HOUSING Equal
Opportunity for All The Fair Housing Act The Fair Housing Act prohibits discrim
```

Figure 12. Sentence-window retrieval output.

4.4 Summary measurements

```
===== SUMMARY =====
[
  {
    "technique": "token",
    "chunks": 88,
    "average_chunk_length": 2290.2,
    "top1_cosine": 0.6283,
    "mean_at_k_cosine": 0.5765,
    "recall_at_k": 1.0,
    "mean_retrieval_latency_ms": 132.392
  },
  {
    "technique": "semantic",
    "chunks": 51,
    "average_chunk_length": 3454.65,
    "top1_cosine": 0.631,
    "mean_at_k_cosine": 0.551,
    "recall_at_k": 0.8,
    "mean_retrieval_latency_ms": 19.03
  },
  {
    "technique": "sentence_window",
    "chunks": 951,
    "average_chunk_length": 185.26,
    "top1_cosine": 0.749,
    "mean_at_k_cosine": 0.6902,
    "recall_at_k": 0.8,
    "mean_retrieval_latency_ms": 50.191
  }
]
```

Figure 13. Retrieval summary written by the experiment.

5. Results and Interpretation

Technique	Chunks	Avg length	Top-1 cosine	Mean@3 cosine	Recall@3	Mean latency (ms)
Token	88	2290.20	0.6283	0.5765	1.0000	132.392
Semantic	51	3454.65	0.6310	0.5510	0.8000	19.030
Sentence-window	951	185.26	0.7490	0.6902	0.8000	50.191

Recall@3 here is source-level recall: it is true when at least one of the top three nodes comes from the expected source document. It is not a claim that every returned chunk contains the exact answer.

The sentence-window configuration produced the strongest similarity values: its top-1 cosine was 0.7490 and its mean cosine across the top three was 0.6902. Semantic retrieval was fastest at 19.030 ms on this run, while token retrieval reached perfect source-level Recall@3 but returned larger chunks and had the highest mean latency. The trade-off is therefore clear: sentence windows improved semantic proximity, semantic chunks minimized lookup time, and token chunks gave the most consistent source coverage.

A useful failure case was the security-deposit query. One high-scoring result discussed a military-service exception rather than the general 21-day return rule. This demonstrates why similarity scores should be inspected with the retrieved preview and source, not treated as proof that the answer is correct.

6. Conclusion

After testing the application and reviewing the retrieval output, I found that the main requirements were working together: the portal protected the dashboard, the session cookie included the required security attributes, and the retrieval script produced comparable machine-readable results. My results suggest that sentence-window retrieval was the best overall match for this corpus, while semantic chunking was the fastest. The security-deposit example also reminded me that a high similarity score is not enough; I still need to inspect the returned passage to decide whether it answers the question.

7. AI Use and Verification

I used an AI assistant mainly to break the assignment requirements into smaller tasks, troubleshoot installation and coding problems, review my experimental setup, and improve the clarity of my report. I completed the implementation, downloaded and selected the corpus documents, configured the environment, ran the retrieval experiments, captured the screenshots, and reviewed the results myself.

One problem occurred during the first retrieval run. The script completed, but several retrieved text previews were unreadable because the PDF text extraction did not preserve the document content correctly. I identified the problem by inspecting the actual retrieved passages instead of relying only on the similarity scores and summary metrics. I then changed the extraction process to use PyMuPDF, limited the document text to a manageable size, and repeated the experiment. The second run produced readable previews and meaningful source comparisons, so I used those repeated results in this report and excluded the earlier faulty output.

Appendix A — Relevant authentication code

This is the complete authentication router used for the login, protected-dashboard, timeout, and logout behavior:

```
import time
from fastapi import APIRouter, Request, Form
from fastapi.responses import RedirectResponse
from fastapi.templating import Jinja2Templates
from starlette.status import HTTP_302_FOUND

router = APIRouter()
```

```

templates = Jinja2Templates(directory="templates")

VALID_USERNAME = "admin"
VALID_PASSWORD = "password"
IDLE_TIMEOUT_SECONDS = 300

def get_active_user(request: Request):
    user = request.session.get("user")
    if not user:
        return None
    last_activity = request.session.get("last_activity", 0)
    if time.time() - last_activity > IDLE_TIMEOUT_SECONDS:
        request.session.clear()
        return None
    request.session["last_activity"] = time.time()
    return user

@router.get("/")
def home(request: Request):
    return templates.TemplateResponse("index.html", {"request": request, "user":
get_active_user(request)})

@router.get("/login")
def login_page(request: Request, error: str = ""):
    return templates.TemplateResponse("login.html", {"request": request, "user":
get_active_user(request), "error": error})

@router.post("/login")
def login(request: Request, username: str = Form(...), password: str = Form(...)):
    if username == VALID_USERNAME and password == VALID_PASSWORD:
        request.session["user"] = username
        request.session["last_activity"] = time.time()
        return RedirectResponse(url="/dashboard", status_code=HTTP_302_FOUND)
    return RedirectResponse(url="/login?error=invalid", status_code=HTTP_302_FOUND)

@router.get("/dashboard")
def dashboard(request: Request):
    user = get_active_user(request)
    if not user:
        return RedirectResponse(url="/login", status_code=HTTP_302_FOUND)
    return templates.TemplateResponse("dashboard.html", {"request": request, "user": user})

@router.get("/logout")
def logout(request: Request):
    request.session.clear()
    return RedirectResponse(url="/", status_code=HTTP_302_FOUND)

```

The same file is stored in the repository at `routers/auth.py` and is included here so the screenshots can be matched directly to the implementation.